Road Sign Classification with Denoising Pipeline Approach

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Abstract: Reliable traffic sign recognition is critical for autonomous vehicles. However, real-world images are affected by noise, which degrades the' quality. This research proposes a denoiser-classifier pipeline approach, which uses a pre-trained color image denoiser, and compares it against an end-to-end approach (single classification models). This paper evaluates the pipeline approach and the singular model's image recognition performance on the German Traffic Sign Recognition Benchmark (GTSRB) dataset under Gaussian noise with standard deviation in pixel value units from 0 to 100. Tests on CNN and ResNet-18 show that the pipeline improves CNN models' performance under moderate noise but offers limited to no gains for ResNet-18, which generally performs better end-to-end. This study provides insights into the importance of model choice for noise-resilient recognition systems.

Keywords— Classification, Pipeline, Noise, Image Processing, Computer Vision.

I. INTRODUCTION

The growing number of self-driving vehicles shows their potential to improve transportation efficiency and convenience, highlighting the necessity to investigate the safety of autonomous driving.

In real-life environments, automated systems often encounter various types of noise due to low-light environments or poor weather conditions. The challenge in producing high-quality images in low-light scenarios is that the signal-to-noise ratio is greatly reduced [1]. This can result in noise becoming the dominant factor in a captured image, thereby making machine vision recognition systems less effective at night. This will significantly affect the safety of automated driving systems, since inaccurate predictions of traffic signs could lead to safety hazards [2]. Therefore, ensuring the accuracy of street sign identification under the influence of noise is critical for the safety of passengers and pedestrians. To address this challenge, we have explored different approaches. One of our methods is the end-to-end approach, which uses noisy data to train a model directly. Although this approach helps the model become naturally robust to noise, it requires a large amount of training data to achieve good performance [3] and generalize well across different noise types and intensities [4].

This research proposes a pipeline approach that uses a weak classifier after a pre-trained denoiser model (PTM). It has multiple advantages, especially in terms of efficiency. For example, the denoiser in this research is a pre-trained model not only for traffic sign image denoising but also for general-colored image denoising. The denoiser-classifier pipeline is especially beneficial when there is a large amount of unlabeled data, since the denoiser can be fine-tuned well in an unsupervised or self-supervised way [5]. However, the pipeline approach is typically slower in inference [6], and the denoiser (without fine-tuning) may reflect human assumptions about noise types or prior biases [7].

The goal of this research is to identify the most effective approaches for use in self-driving vehicles and to investigate the impact of various training data and model architectures on classification accuracy. The research will compare the pipeline approach with the end-to-end approach. The best method will be judged based on the ability to maintain high accuracy in noisy environments. This research will compare the pipeline approach's and end-to-end approaches' performances, depending on the noise level and neural network architecture.

II. METHODOLOGY

We consider the multi-class classification setting and denote by h_c the classifier trained on clean data, by h_n^i the classifier trained on noisy data. The noise is Gaussian noise [8] in our setting, and i in h_n^i denotes the noise level, which is the standard deviation of the added noise. Consider X as the input space and Y as the set of 43 classes. Then $h: X \times Y \rightarrow R$ and $h(x) = argmax_{\{y \in Y\}} h(x, y)$.

We denote by d the color image denoiser. For the pipeline classifier composed of a denoiser and a classifier, it is denoted as $p_{h_n^i}$.

$$p_{h_n^i} = h_n^i \circ d$$

$$p_c = h_c \circ d$$

We will show that while both $p_{h_n^i}$, p_c, and h_n^i show some robustness to Gaussian noise, compared to h_c. However, their performance depends on the noise level and the neural network architecture.

III. Theoretical Analysis

In the denoiser-classifier pipeline approach, the classifier can operate on less corrupted representations. Here, a well-trained denoiser can be reused across tasks or models. This pipeline approach also enhances interpretability, as it is easier to locate whether there are issues with denoising or classification.

Additionally, the denoiser-classifier pipeline is particularly beneficial when unlabeled data is abundant but labeled data is scarce [5]. Researchers proposed a pipeline approach, which leverages a pre-trained denoising model, and demonstrated that it improves classification accuracy in noisy environments by effectively preprocessing data [9]. Without clean data, a simple general-purpose neural network can also achieve high restoration performance of a deep neural network [10].

In assumption, the compound model achieves its best performance when the classifier's training set and the denoiser's output set are consistent. However, the pipeline approach, like most two-stage processes, is typically slower to operate [11]. Also, the denoiser often reflects human assumptions about noise types or priors.

In contrast, the classifier trained end-to-end on noisy data without an explicit denoising step is simpler and has a faster inference time. End-to-end optimization helps the model to be robust to noise naturally, without relying on handcrafted priors [12], but this end-to-end approach may require large and diverse training data to generalize well across noise types and intensities [13].

IV. Experiment

A. Dataset

We use GermanTrafficSignRecognitionBenchmark dataset [14]. The GTSRB has more than 43 classes and more than 50,000 images in total.

Our research used GTSRB_Final_Training_Images for training [14] and GTSRB_Final_Test_GT for testing [14].

B. Noise Level

Noise level quantifies the standard deviation of the Probability Density Function (PDF) of the Gaussian noise, which is in the range from 0 to 100, increasing by 20 each time. This produces 6 testing sets with 0, 20, 40, 60, 80, and 100 levels of Gaussian noise.

For this study, the range of noise levels from 0 to 100 is chosen because it starts at 0, where the image is the original (without noise), which acts as a baseline for later evaluations, to 100, where the artificially induced noise is obvious and the effect on image quality is clearly visible. As the noise level increases above 100, the induced noise becomes too great for any meaningful classification by the models; thus, it will not be investigated in this study. This range is considered reasonable and offers diverse images to be tested.

C. Classifier

The CNN we use has total parameters: 2,201,707 (Trainable parameters: 2,201,707; Non-trainable parameters: 0). The ResNet-18 we use has total parameters: 11,198,571 (Trainable parameters: 11,198,571; Non-trainable parameters: 0).

When training, we set the number of epochs as 5, by which time the model has converged and has not overfitted.

D. Implementation

Section 1 - Preparing the Dataset

1. Download the training and testing dataset from GTSRB
2. Duplicate the training set and add Gaussian noise with a level of 20.
3. Repeat the step above with a noise level of 60. Save 2 noisy training sets.
4. Duplicate the testing set and use the script to add a Gaussian noise level of 20 to it.
5. Repeat the step above with noise levels of 40, 60, 80, 100. Save 5 noisy testing set.

Section 2 - Creating and Training the Models

1. Set up the CNN training loop and train it with the clean dataset, save the model
2. Train CNN with the noisy datasets (with noise level 20 and 60), save the models
3. Import the ResNet-18 and train it with the clean dataset, save the model.
4. Train ResNet-18 with the noisy datasets (with noise levels 20 and 60), save the models
5. Employ pre-trained SwinIR (colorDN_SwinIRM_noise15) to create denoised images (predenoised noise level being 20, 40, 60, 80, 100) and store denoised images. *Results: have the 6 models(4 end-to-end and 2 compound models) and 5 denoised testing image sets ready.*

Section 3 - Testing

1. Load the 6 models and test their performances on testing sets with different levels of noise.
2. Load the 6 models and test their performances on denoised testing sets with different levels of noise.

V. Result Analysis

The test accuracy on various testing sets is given in Table 1 and Table 2.

Fig. 1a shows that ResNet trained on clean data is not robust to Gaussian noise, with its accuracy monotonically decreasing with noise level. ResNet trained on noisy data exhibits robustness to Gaussian noise. Both ResNets trained with 20 levels of Gaussian noise and trained with 60 levels of Gaussian noise have their accuracy increase and then decrease as the noise level in the testing set increases. Interestingly, two curves are quite similar, reaching there highest accuracy with testing noise level around 40.

Table 1. Accuracy of CNNs

TESTING NOISE LEVEL	CLEAN	DIRTY 20	DIRTY 60	PIPE C	PIPE N20	PIPE N60
0	0.5587	0.5010	0.5012	0.5587	0.5010	0.5012
20	0.5028	0.5065	0.5177	0.5100	0.6310	0.4870
40	0.4317	0.4899	0.5093	0.4417	0.6802	0.4995
60	0.3731	0.4569	0.4751	0.3798	0.6411	0.4822
80	0.3298	0.4017	0.4191	0.3285	0.5755	0.4203
100	0.2896	0.3558	0.3667	0.2894	0.3576	0.3721

Table 2. Accuracy of ResNets

TESING NOISE LEVEL	CLEAN	DIRTY 20	DIRTY 60	PIPE C	PIPE N20	PIPE N60
0	0.9268	0.6344	0.5341	0.9268	0.6344	0.5341
20	0.6736	0.7232	0.6953	0.8039	0.4890	0.5595
40	0.4013	0.7013	0.7064	0.4490	0.4819	0.6871
60	0.2328	0.6461	0.6627	0.2618	0.4611	0.6592
80	0.1349	0.5750	0.5907	0.1520	0.4110	0.5907
100	0.0895	0.4979	0.5134	0.0964	0.5044	0.5173

For the pipeline approaches with ResNet18 (Fig. 1b), the situation is similar to the end-to-end approach (Fig. 1a). Denoiser does not seem to improve accuracy when connected with non-robust ResNet trained on clean data. For ResNets trained on noisy data, when combined with a denoiser, they still outperform ResNets trained on clean data combined with a denoiser and can maintain a stable accuracy without performance degradation as the noise level increases.

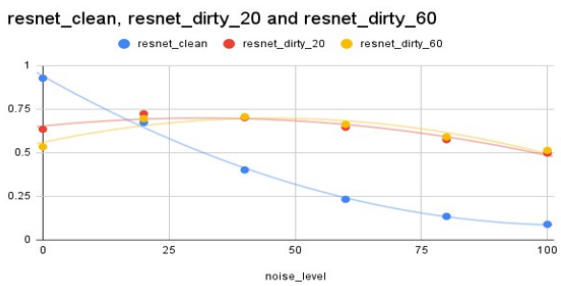
Although in the test which has high-level noise, models trained on noisy data outperform models trained on clean data, when testing on clean data, models trained on noisy data cannot do better than its counterpart, which is trained on clean data.

In CNN, a similar result is shown in Fig. 2.

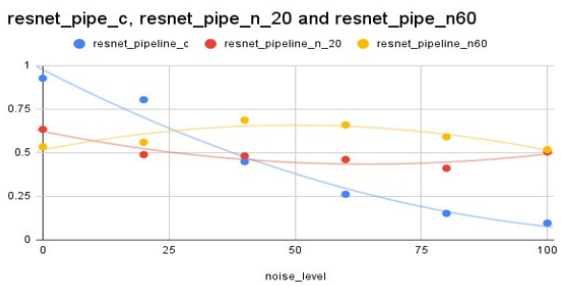
The CNN that has the best performance on noisy data is cnn_pipeline_n20, that is, the pipeline approach with CNN trained on noisy data with noise level 20. The ResNet that has the best performance on noisy data is resnet_dirty_20, resnet_dirty_60, resnet_pipeline_n60, that is, the ResNet trained on noisy data with noise level 20, the ResNet trained on noisy data with noise level 60, and the pipeline approach with ResNet trained on noisy data with noise level 60. All 4 of them achieve their highest accuracy of about 70% on moderately noisy data.

When we compare the end-to-end approach and the pipeline approach, Fig. 3 shows that on CNNs trained with clean or 60-level noise data, the denoiser does not change the accuracy. There is neither much benefit nor penalty for using the pipeline approach. However, when CNN is trained on noisy data with a noisy level equal to 20, the pipeline approach outperforms the end-to-end approach when the testing noise is moderate.

On ResNet18, the situation is reversed. The end-to-end approach outperforms the pipeline approach when the training noise level equals 20.

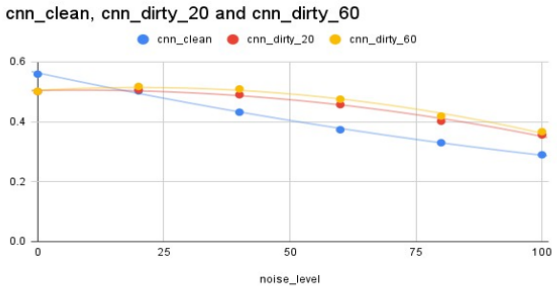


(a) Single ResNet

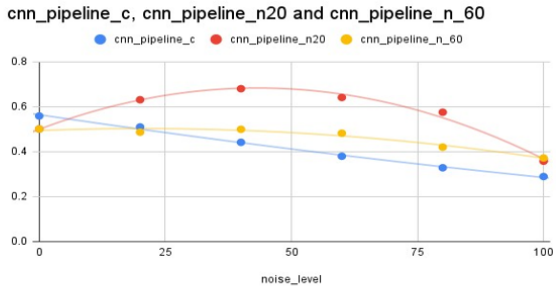


(b) Pipeline ResNet

Figure 1: ResNet by Noise Level

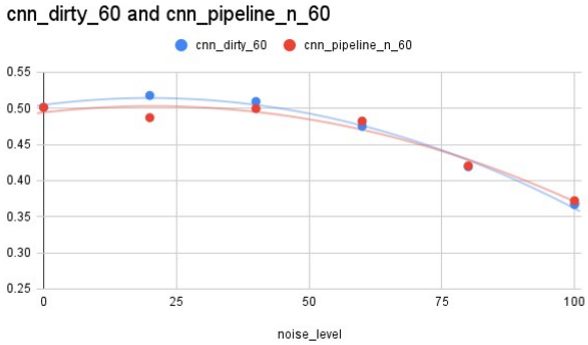


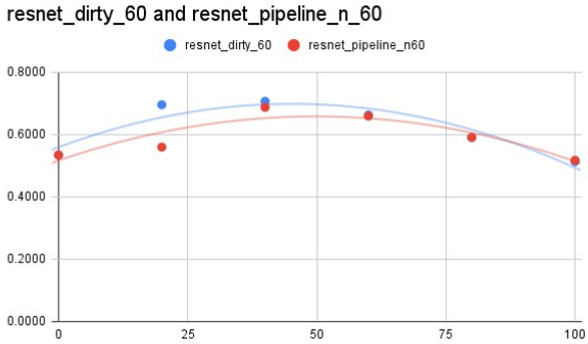
(a) Single CNN

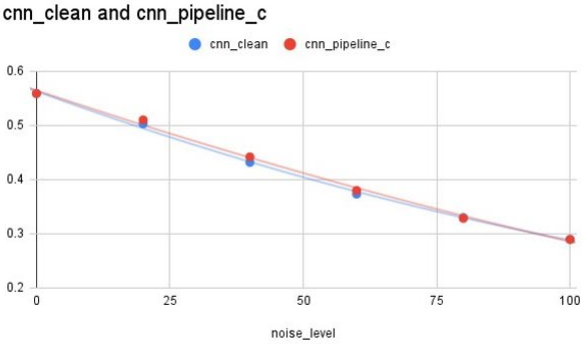


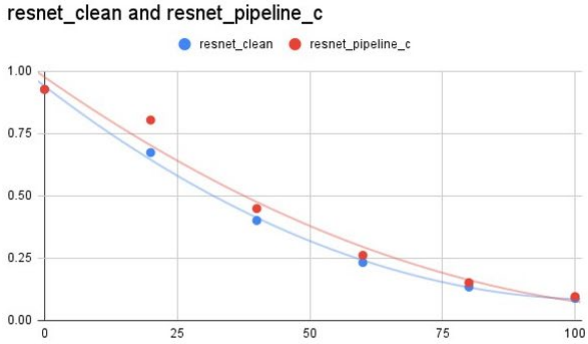
(b) Pipeline CNN

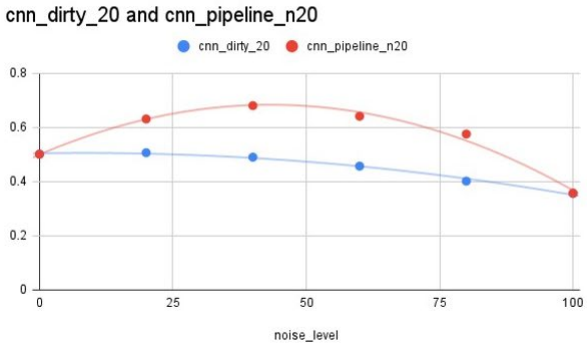
Figure 2: CNN by Noise Level CNN - Clean ResNet - Clean











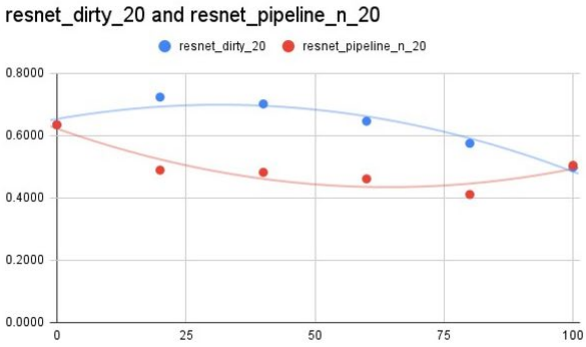


Figure 3. Comparison of Single Model and Pipeline

VI. Limitation

A. Dataset

The GSTRB, despite being a benchmark in the field of traffic sign classification, has an unbalanced number of images in different categories. As shown in Figure 4, while certain traffic sign classes are represented by thousands of training samples, others are limited to just a few hundred.

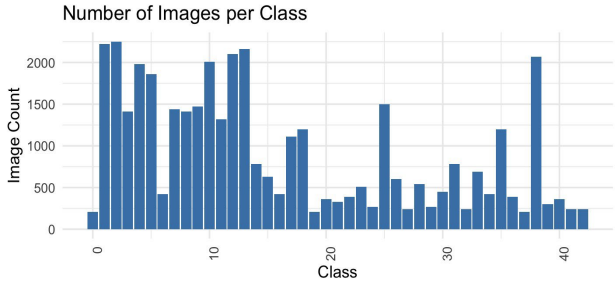


Figure 4: number of images in different classes

This majority class bias, which is that the model may learn to always predict the majority class, achieving high accuracy by ignoring minority classes [15]. As a result, the overall accuracy might appear high, while the model underperforms on minority classes, failing to recognize less frequent yet equally important signs. This limitation can hinder the model's generalization ability, especially when deployed in environments where rare traffic signs are critical for safety and compliance.

B. Noise

In real life, Gaussian noise is not always the best way to describe the kind of noise you'd see in a noisy environment. In practical scenarios, faulty or aging camera sensors can result in dead or stuck pixels, producing Salt-and-Pepper noise [16], which exhibits sharp pixel-level distortions. Moreover, the randomness in how photons hit the camera sensor during image capture, especially in low-light conditions, leads to signal-dependent noise patterns. [17].

Besides, the range of noise levels may be reconsidered. Although noise levels up to 100 were introduced to systematically evaluate the model's robustness under varying conditions, such extreme noise is rarely encountered in real-world scenarios. In practice, images captured by traffic cameras or autonomous vehicles are unlikely to exhibit such a high level of degradation. Noise level 100 often results in severely distorted images that are nearly unrecognizable and may not accurately represent practical conditions [18]. To overcome the identified challenges, there is still much to be done in the future. It would be valuable to integrate a recognition model that can detect and localize the regions of street signs in full street views. The model will automatically detect and mark relevant areas, followed by image cropping to isolate street signs for improved classification performance.

In addition, to improve the accuracy of traffic sign recognition in complex real-world environments, researchers are encouraged to attempt to use deep learning-based image generation models. These models can complete the content of incomplete traffic signs due to graffiti, damage, or obstruction. This approach effectively addresses issues such as malicious damage and natural wear and tear, providing greater practicality for traffic sign recognition systems under harsh conditions [19].

Also, instead of using the pre-trained model directly, it is promising to fine-tune the model based on traffic sign images by transfer learning.

VII. Conclusion

This research presented a detailed analysis of the properties of the pipeline approach in road sign image denoising. It demonstrates that with a classifier trained on a moderate level of noise, the accuracy of the pipeline approach may be different from the end-to-end approach. The pipeline approach can be beneficial in CNN and also may be harmful in ResNet-18.

Overall, our findings demonstrate the importance of matching the denoising model with the proper classifier architecture and training data. These results can inform the design of image recognition systems in autonomous vehicles, which is critical under real-world noise conditions.

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